

10. Fluorine exists naturally as the fluoride ion. It is found in soil, water, foods and several minerals, such as fluorapatite and fluorite.

Fluoride ion concentration in seawater averages 1.3 ppm (parts per million). In fresh water, the natural range is typically between 0.01 and 0.3 ppm. In some parts of the world, fresh water contains fluoride ion levels which are dangerous and can lead to health problems.

In the early 1930s, scientists found that people who were brought up in areas with naturally fluoridated water had up to two-thirds fewer cavities compared to those who lived in areas where the water was not fluoridated. Several studies since then have repeatedly shown that when fluoride is added to people's drinking water in areas where natural levels are low, tooth decay decreases.

However, many European countries which do not fluoridate their water do not have a higher incidence of dental decay than countries which do so. It was also found that in Germany and Finland, decay rates either remained stable or continued in their downward trend after they stopped adding fluoride to their drinking water.

Figure 1 shows data about the effect of fluoridation of drinking water on the mean number of decayed, missing and filled teeth (DMFT) and the amount of fluorosis seen.

Figure 2 shows the change in mean DMFT in three regions of Australia over a four year period.

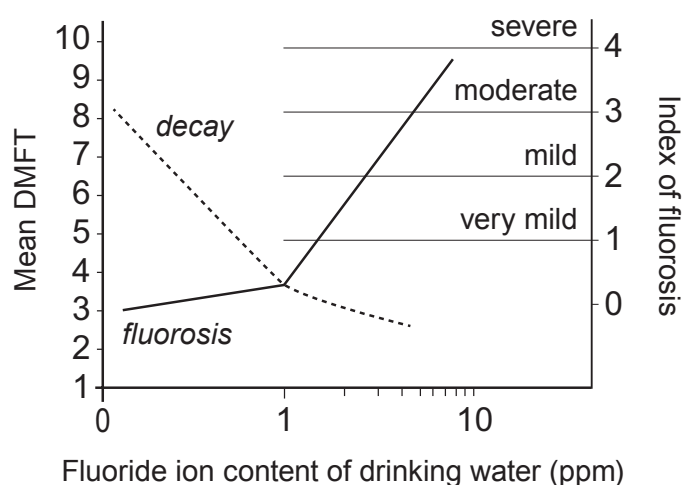


Figure 1



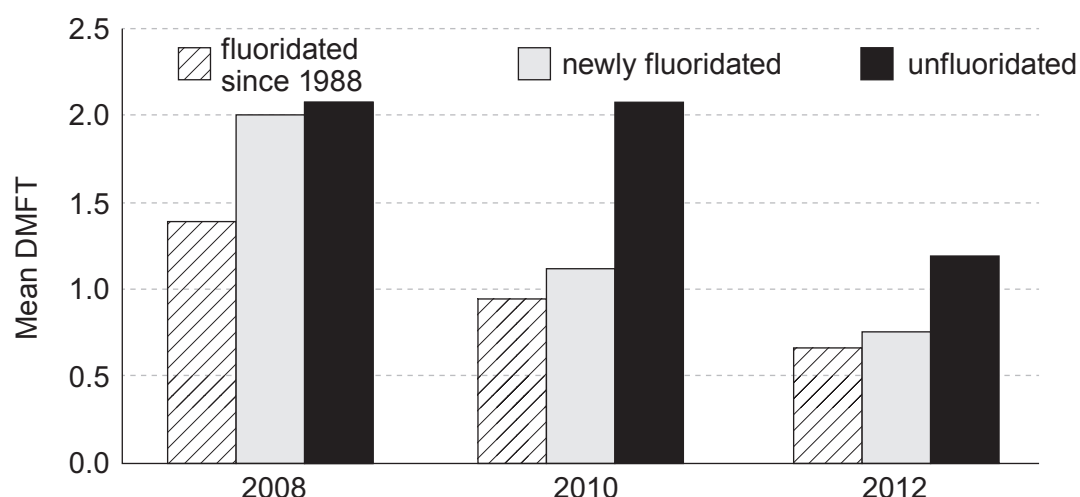


Figure 2

- (a) Describe the effects of adding varying concentrations of fluoride ions to drinking water. [3]

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- (b) Tick (✓) the statement below that best describes why Germany stopped fluoridating its water supplies. [1]

Ten years after stopping adding fluoride there was no increase in tooth decay

☐

They found that adding fluoride caused fluorosis

☐

Natural water supplies already contain fluoride in a high concentration

☐

Studies showed that areas with no fluoridation did not have higher levels of decay than areas that did fluoridate

☐


- (c) Tick (✓) the box which gives **one** definite conclusion that can be drawn using **only** the data in **Figure 2**. [1]

Fluoridation has no effect on levels of decay

☐

People have reduced their intake of sugary foods over this period

☐

More than one factor affects levels of decay

☐

Fluoridation is the main cause of falling levels of decay

☐

- (d) 'Mass medication' is an argument often given to oppose fluoridation of water supplies. Explain what is meant by the term *mass medication*. [1]

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END OF PAPER

